

Solutions to Norris' **Markov Chains**

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June 2026

I was working through *Markov Chains*, Cambridge University Press, 1997 by J.R. Norris and decided that I might as well provide my solutions in the off-chance that they are of use to others. If interested, see olivermaynard.com for more about me and the things that I do.

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1 Discrete-time Markov chains

1.1 Definition and Basic Properties

Exercise 1.1.1. Let $B_1, B_2, \dots$ be disjoint events with $\bigcup_{n=1}^{\infty} B_n = \Omega$. Suppose A is another event with $\mathbb{P}(A \mid B_n) = p$ for every n . Show that $\mathbb{P}(A) = p$. Deduce that if X and Y are discrete random variables, then the following are equivalent:

- (a) X and Y are independent;
- (b) the conditional distribution of X given $Y = y$ is independent of y .

Proof. Consider first that

$$\mathbb{P}(A \mid B_n) = p \quad \text{for } n = 1, 2, \dots,$$

and that by the law of total probability,

$$\mathbb{P}(A) = \sum_{n=1}^{\infty} \mathbb{P}(A \mid B_n) \mathbb{P}(B_n) = \sum_{n=1}^{\infty} p \mathbb{P}(B_n).$$

Since the B_n are disjoint with $\bigcup_{n=1}^{\infty} B_n = \Omega$, we have $\sum_{n=1}^{\infty} \mathbb{P}(B_n) = \mathbb{P}(\Omega) = 1$, and

$$\sum_{n=1}^{\infty} p \mathbb{P}(B_n) = p \sum_{n=1}^{\infty} \mathbb{P}(B_n) = p(1) = p.$$

So

$$\mathbb{P}(A) = p.$$

□

We now demonstrate the equivalence of (a) and (b).

(a) $\Rightarrow$ (b). If X and Y are independent, then

$$\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x) \mathbb{P}(Y = y) \quad \text{for all } x, y.$$

Now consider that, as a direct result,

$$\mathbb{P}(X = x \mid Y = y) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)} = \mathbb{P}(X = x).$$

Since this is true for every value x and every y , the conditional distribution of X given $Y = y$ is, for each y , exactly the distribution of X ; i.e.,

$$\mathbb{P}(X \mid Y = y) = \mathbb{P}(X) \quad \text{for all } y,$$

which is statement (b).

(b) $\Rightarrow$ (a). If the conditional distribution of X given $Y = y$ is independent of y , then

$$\mathbb{P}(X \mid Y = y) = \mu(X) \quad \text{for all } y \text{ (with } \mathbb{P}(Y = y) > 0),$$

where $\mu(X)$ is some generic distribution. Since $\mu(X)$ is constant for all y , we may use the above result regarding $\mathbb{P}(A \mid B_n) = p$ and obtain

$$\mathbb{P}(X) = \mu(X) \implies \mathbb{P}(X \mid Y = y) = \mathbb{P}(X).$$

So

$$\underbrace{\mathbb{P}(X = x \mid Y = y)}_{= \mathbb{P}(X=x)} = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)} \implies \mathbb{P}(X = x) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)}.$$

Therefore,

$$\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x) \mathbb{P}(Y = y) \quad \text{for all } x, y,$$

which is exactly the definition of independence between X and Y .

Exercise 1.1.2. Suppose that $(X_n)_{n \geq 0}$ is Markov(λ, P). If $Y_n = X_{kn}$, show that $(Y_n)_{n \geq 0}$ is Markov(λ, P^k).

Proof. Since $Y_n = X_{kn}$, we observe that

$$\mathbb{P}(Y_0 = i_0, \dots, Y_n = i_n) = \mathbb{P}(X_0 = i_0, X_k = i_1, \dots, X_{kn} = i_n).$$

By the Markov property, this probability factors as

$$\mathbb{P}(X_0 = i_0, X_k = i_1, \dots, X_{kn} = i_n) = \lambda_{i_0} p_{i_0 i_1}^* p_{i_1 i_2}^* \cdots p_{i_{n-1} i_n}^*,$$

where the p_{ij}^* are the transition probabilities of the chain $(X_n)_{n \geq 0}$ from a state i at X_m (for any integer $m \geq 0$) to a state j at X_{m+k} , i.e., k steps of the chain later. Formally, this probability is

$$p_{ij}^* = \mathbb{P}_i(X_{m+k} = j) = \mathbb{P}(X_{m+k} = j \mid X_m = i),$$

which was shown in Theorem 1.1.3(ii) to be exactly $p_{ij}^{(k)}$, the (i, j) entry in P^k . So

$$\mathbb{P}(Y_0 = i_0, \dots, Y_n = i_n) = \lambda_{i_0} p_{i_0 i_1}^{(k)} p_{i_1 i_2}^{(k)} \cdots p_{i_{n-1} i_n}^{(k)},$$

and by Theorem 1.1.1, this is exactly the statement that $(Y_n)_{n \geq 0}$ is Markov(λ, P^k). This result makes intuitive sense, since we may interpret each step and transition probability between states in the chain $(Y_n)_{n \geq 0}$ as corresponding to a k -step journey between the same states in $(X_n)_{n \geq 0}$. $\square$

Exercise 1.1.3. Let X_0 be a random variable with values in a countable set I . Let $Y_1, Y_2, \dots$ be a sequence of independent random variables, uniformly distributed on $[0, 1]$. Suppose we are given a function

$$G : I \times [0, 1] \rightarrow I$$

and define inductively

$$X_{n+1} = G(X_n, Y_{n+1}).$$

Show that $(X_n)_{n \geq 0}$ is a Markov chain and express its transition matrix P in terms of G . Can all Markov chains be realized in this way? How would you simulate a Markov chain using a computer?

Suppose now that $Z_0, Z_1, \dots$ are independent, identically distributed random variables such that $Z_i = 1$ with probability p and $Z_i = 0$ with probability $1 - p$. Set $S_0 = 0$, $S_n = Z_1 + \cdots + Z_n$. In each of the following cases, determine whether $(X_n)_{n \geq 0}$ is a Markov chain:

- (a) $X_n = Z_n$;
- (b) $X_n = S_n$;
- (c) $X_n = S_0 + \cdots + S_n$;
- (d) $X_n = (S_n, S_0 + \cdots + S_n)$.

In the cases where $(X_n)_{n \geq 0}$ is a Markov chain, find its state-space and transition matrix; in the cases where it is not, give an example where $\mathbb{P}(X_{n+1} = i \mid X_n = j, X_{n-1} = k)$ is not independent of k .

Proof. We first consider the probability of the intersection

$$\mathbb{P}(X_0 = i_0, \dots, X_n = i_n) = \mathbb{P}(X_0 = i_0) \cdot \mathbb{P}(X_1 = i_1 \mid X_0 = i_0) \cdots \mathbb{P}(X_n = i_n \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1})$$

where the conditional probabilities are each given by (using an arbitrary index m in the chain)

$$\mathbb{P}(X_{m+1} = i_{m+1} \mid X_0 = i_0, \dots, X_m = i_m) = \mathbb{P}(G(X_m, Y_{m+1}) = i_{m+1} \mid X_0 = i_0, \dots, X_m = i_m).$$

Since $X_0, \dots, X_m$ are determined entirely by X_0 and $Y_1, \dots, Y_m$, and Y_{m+1} is independent of $Y_1, \dots, Y_m$ (and of X_0), it follows that Y_{m+1} is independent of $X_0, \dots, X_m$. Thus

$$\mathbb{P}(G(X_m, Y_{m+1}) = i_{m+1} \mid X_0 = i_0, \dots, X_m = i_m) = \mathbb{P}(G(i_m, Y_{m+1}) = i_{m+1}).$$

Further, observe that since $Y_1, Y_2, Y_3, \dots$ are independent and identically distributed uniformly in $[0, 1]$, $G(i, \cdot)$ is computed the same regardless of which Y_{m+1} you plug in (i.e., which time index you happen to be

looking at). For any transition from state i to j this means we have

$$\mathbb{P}(G(i, Y_{m+1}) = j) = \mathbb{P}(G(i, Y) = j) \text{ where } Y \sim \text{Uniform}[0, 1],$$

and can then define the transition probabilities in the natural way

$$p_{ij} := \mathbb{P}(G(i, Y) = j).$$

We can finally write

$$\mathbb{P}(X_0 = i_0, \dots, X_n = i_n) = \lambda_{i_0} p_{i_0 i_1} p_{i_1 i_2} \cdots p_{i_{n-1} i_n},$$

and by Theorem 1.1.1 conclude that $(X_n)_{n \geq 0}$ with $X_0 \sim \lambda$ is therefore Markov(λ, P), with the (i, j) entry of the transition matrix given by the corresponding transition probability p_{ij} . $\square$

Remarkably, *any* Markov chain can be realized in this way. To see this, fix a state i . The row $(p_{ij})_{j \in I}$ of P is a probability distribution over I . Since I is countable, we can list its elements $j_1, j_2, j_3, \dots$ and partition $[0, 1]$ into consecutive intervals

$$A_{ij_1} = [0, p_{ij_1}), \quad A_{ij_2} = [p_{ij_1}, p_{ij_1} + p_{ij_2}), \quad A_{ij_3} = [p_{ij_1} + p_{ij_2}, p_{ij_1} + p_{ij_2} + p_{ij_3}), \dots$$

each having exactly length p_{ij_k} . We then define

$$G(i, u) = j_k \quad \text{whenever } u \in A_{ij_k}.$$

Observe that for U uniform on $[0, 1]$,

$$\mathbb{P}(G(i, U) = j_k) = \mathbb{P}(U \in A_{ij_k}) = \text{length of } A_{ij_k} = p_{ij_k}$$

which exactly matches the prescribed transition probability. So the chain $X_{n+1} = G(X_n, Y_{n+1})$ built from this G has exactly transition matrix P (*any* P you started with). This construction is exactly the way you would simulate a Markov chain on a computer. Given you're in state i , draw a uniform random number U and use the cumulative sums of the row p_i to decide which "bucket" U falls into (which tells you the next state). The G function is in essence just the inverse cumulative density function of row i of P .

Classifying Random Processes:

(a) We consider $X_n = Z_n$ and

$$\begin{aligned} \mathbb{P}(X_0 = i_0, \dots, X_n = i_n) &= \mathbb{P}(X_0 = i_0) \cdot \mathbb{P}(X_1 = i_1 \mid X_0 = i_0) \cdots \\ &\quad \mathbb{P}(X_n = i_n \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}). \end{aligned}$$

Since $X_n = Z_n$, and $Z_0, Z_1, \dots$ are i.i.d., X_n is independent of $X_0, \dots, X_{n-1}$ entirely. In particular,

$$\mathbb{P}(X_n = i_n \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}) = \mathbb{P}(X_n = i_n) = \mathbb{P}(Z_n = i_n),$$

which satisfies the (trivial) Markov property

$$\mathbb{P}(X_n = i_n \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}) = \mathbb{P}(X_n = i_n \mid X_{n-1} = i_{n-1}),$$

since neither side depends on any of $i_0, \dots, i_{n-2}$, or even on i_{n-1} . Substituting this into the original factorization gives

$$\mathbb{P}(X_0 = i_0, \dots, X_n = i_n) = \lambda_{i_0} p_{i_0 i_1} \cdots p_{i_{n-1} i_n},$$

where $p_{ij} := \mathbb{P}(Z = j)$ for any i , and thus $(X_n)_{n \geq 0}$ is Markov(λ, P). Z only takes on values 0 and 1, so the state-space I is just the set $\{0, 1\}$, with the transition matrix and initial distribution given directly by the problem:

$$P = \begin{pmatrix} p & 1-p \\ p & 1-p \end{pmatrix}, \quad \lambda = \begin{pmatrix} p \\ 1-p \end{pmatrix}.$$

Observe that the rows of P are identical, since the transition probability from *any* state i to state j does not depend on i .

(b) We now take under consideration $X_n = S_n = Z_1 + \cdots + Z_n$ with $X_0 = S_0 = 0$, and observe for a general index in the chain

$$\mathbb{P}(X_n = i_n \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}).$$

Conditioning on $X_0 = 0, X_1 = i_1, \dots, X_{n-1} = i_{n-1}$ is equivalent to conditioning on the individual values $Z_1 = i_1, Z_2 = i_2 - i_1, \dots, Z_{n-1} = i_{n-1} - i_{n-2}$, since each X_m determines Z_m once X_{m-1} is also known. Writing $X_n = X_{n-1} + Z_n = i_{n-1} + Z_n$, we have

$$\mathbb{P}(X_n = i_n \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}) = \mathbb{P}(i_{n-1} + Z_n = i_n \mid Z_1 = i_1, \dots, Z_{n-1} = i_{n-1} - i_{n-2}).$$

Since $Z_1, Z_2, \dots$ are i.i.d., Z_n is independent of $Z_1, \dots, Z_{n-1}$, so conditioning on the earlier Z 's gives no information about Z_n , and the conditional collapses to

$$\mathbb{P}(X_n = i_n \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}) = \mathbb{P}(i_{n-1} + Z_n = i_n) = \mathbb{P}(Z_n = i_n - i_{n-1}) =: p_{i_{n-1}i_n}.$$

This depends only on i_{n-1} and i_n , not on $i_0, \dots, i_{n-2}$, so

$$\mathbb{P}(X_n = i_n \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}) = \mathbb{P}(X_n = i_n \mid X_{n-1} = i_{n-1}) = p_{i_{n-1}i_n},$$

which is exactly the Markov property. Thus,

$$\mathbb{P}(X_0 = i_0, \dots, X_n = i_n) = \lambda_{i_0} p_{i_0i_1} \cdots p_{i_{n-1}i_n},$$

and $(X_n)_{n \geq 0}$ is a Markov chain. Since $X_n = Z_1 + \cdots + Z_n$ and each $Z_i \in \{0, 1\}$, the sum can take any non-negative integer value, increasing by at most 1 each step starting from $X_0 = 0$. So $I = \{0, 1, 2, \dots\} = \mathbb{Z}_{\geq 0}$, the non-negative integers. The transition matrix follows from the definition of each $p_{i_{n-1}i_n}$:

$$p_{ij} = \begin{cases} 1-p & \text{if } j = i, \\ p & \text{if } j = i+1, \\ 0 & \text{otherwise.} \end{cases}$$

- (c) The chain $X_n = S_0 + \cdots + S_n$ with $X_0 = S_0 = 0$ is not a Markov chain. We construct an example where $\mathbb{P}(X_{n+1} = i \mid X_n = j, X_{n-1} = k)$ is not independent of k . Consider two paths leading to the same value ($j = 3$) of X_3 (i.e. $n = 3$):

- Path $(Z_1, Z_2, Z_3) = (1, 0, 0)$ gives $X_2 = S_1 + S_2 = 2$ and $X_3 = X_2 + S_3 = 3$, so $k = X_2 = 2$.
- Path $(Z_1, Z_2, Z_3) = (0, 1, 1)$ gives $X_2 = S_1 + S_2 = 1$ and $X_3 = X_2 + S_3 = 3$, so $k = X_2 = 1$.

Considering $X_4 = 5$ (a value reachable from both branches, via opposite outcomes of Z_4) in the former case

$$\mathbb{P}(X_4 = 5 \mid X_3 = 3, X_2 = 2) = \mathbb{P}(Z_4 = 1) = p,$$

since this path forces $S_3 = 1$, giving $X_4 = 4 + Z_4$, so $X_4 = 5 \Leftrightarrow Z_4 = 1$. In the latter case,

$$\mathbb{P}(X_4 = 5 \mid X_3 = 3, X_2 = 1) = \mathbb{P}(Z_4 = 0) = 1 - p,$$

since this path forces $S_3 = 2$, giving $X_4 = 5 + Z_4$, so $X_4 = 5 \Leftrightarrow Z_4 = 0$. These two conditional probabilities differ whenever $p \neq \frac{1}{2}$, so $\mathbb{P}(X_{n+1} = i \mid X_n = j, X_{n-1} = k)$ is not independent of k , contradicting the Markov property. Hence $(X_n)_{n \geq 0}$ is not a Markov chain.

- (d) Recall that this issue with the chain in problem (c) was that $X_n = S_0 + \cdots + S_n$ alone doesn't tell you S_n specifically, and so different paths can give the same cumulative sum but different "current" S_n (which informs the next increment).

Here, we have a pair $X_n = (S_n, S_0 + \cdots + S_n)$. We can write $X_n = (S_n, T_n)$ where $T_n := S_0 + \cdots + S_n$. Then

$$S_{n+1} = S_n + Z_{n+1}, \quad T_{n+1} = T_n + S_{n+1} = T_n + S_n + Z_{n+1}.$$

So $X_{n+1} = (S_{n+1}, T_{n+1}) = (S_n + Z_{n+1}, T_n + S_n + Z_{n+1})$, a function of X_n and the fresh randomness Z_{n+1} alone (exactly the structure of the general $X_{n+1} = G(X_n, Y_{n+1})$ with i.i.d. driving the randomness we saw before!), so the chain satisfies the Markov property, and therefore *is* Markov. The state-space is

$$I = \{(s, t) \in \mathbb{Z}_{\geq 0} \times \mathbb{Z}_{\geq 0} : t \geq s\}.$$

The probabilities of transition $G((s, t), z) = (s + z, t + s + z)$ with $Z \in \{0, 1\}$ are

$$p_{(s,t),(s',t')} = \begin{cases} 1-p & \text{if } (s', t') = (s, t+s), \\ p & \text{if } (s', t') = (s+1, t+s+1), \\ 0 & \text{otherwise.} \end{cases}$$

Exercise 1.1.4 (1.1.4). *A flea hops about at random on the vertices of a triangle, with all jumps equally likely. Find the probability that after n hops the flea is back where it started.*

A second flea also hops about on the vertices of a triangle, but this flea is twice as likely to jump clockwise as anticlockwise. What is the probability that after n hops this second flea is back where it started? [Recall that $e^{\pm i\pi/6} = \sqrt{3}/2 \pm i/2$.]

The First Flea The first flea cannot hop to the vertex it is currently on, and there are three vertices in total, so the transition matrix is

$$P = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}.$$

The probability that after n hops the flea is back where it started is the probability that, conditioned on being at a vertex, the flea makes it back to that vertex after n steps. Since the flea starts deterministically at some vertex i , this is

$$\mathbb{P}_i(X_n = i) = \mathbb{P}(X_{n+m} = i \mid X_m = i) = p_{ii}^{(n)} =: p^{(n)},$$

where by the symmetry of the triangle this value does not depend on which vertex i is, so we drop the subscript. After 0 hops the flea is (trivially) at the vertex where it started, so

$$p^{(0)} = 1.$$

The probability that after n hops the flea is at either other vertex is

$$\frac{1 - p^{(n)}}{2}$$

since the total probability must be equal to 1, and being at either *other* vertex is equally likely. We use the relation $P^{n+1} = P^n P$ to write the generic recurrence

$$p^{(n+1)} = p_{ii}^{(n+1)} = \sum_k p_{ik}^{(n)} p_{ki}.$$

Again, since the flea cannot hop to the vertex where it already is, $p_{ii} = 0$ and

$$p^{(n+1)} = \underbrace{p_{ii}^{(n)} \cdot 0}_{\text{can't jump from } i \text{ to } i} + 2 \cdot \frac{1 - p^{(n)}}{2} \cdot \frac{1}{2} = \frac{1 - p^{(n)}}{2}.$$

This is a standard linear recurrence relation of the form $p^{(n+1)} = ap^{(n)} + b$ with $a = -\frac{1}{2}$ and $b = \frac{1}{2}$. Setting $p^{(n+1)} = p^{(n)} = p^*$ to find the fixed point yields

$$p^* = \frac{1 - p^*}{2} \Rightarrow p^* = \frac{1}{3}.$$

We then write $p^{(n)} = p^* + c \cdot a^n$ (general solution shape of standard linear recurrence) and use the initial condition to determine c :

$$p^{(0)} = \frac{1}{3} + c \cdot \left(-\frac{1}{2}\right)^0 = 1 \Rightarrow c = \frac{2}{3},$$

so

$$p^{(n)} = \frac{1}{3} + \frac{2}{3} \left(-\frac{1}{2}\right)^n.$$

Regarding the Second Flea Unlike its counterpart, the second flea is not equally likely to hop to the other vertices and instead prefers to jump clockwise (it being twice as likely). Thus the transition matrix in this case is

$$P = \begin{bmatrix} 0 & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & 0 \end{bmatrix}.$$

We determine the eigenvalues of P by taking $\det[P - \lambda I] = 0$ ¹:

$$\det \begin{bmatrix} -\lambda & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & -\lambda & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & -\lambda \end{bmatrix} = 0 \Rightarrow \lambda_1 = 1, \lambda_2 = \frac{1}{6}(-3 + i\sqrt{3}), \lambda_3 = \frac{1}{6}(-3 - i\sqrt{3}).$$

¹Alternatively, since P is actually a circulant matrix, the eigenvalues have a known formula!

As stated in **1.1 Definition and Basic Properties**, since the eigenvalues are distinct $p_{ii}^{(n)}$ has the general form

$$p_{ii}^{(n)} = a_1 \lambda_1^n + a_2 \lambda_2^n + a_3 \lambda_3^n.$$

However, since two of the eigenvalues (namely λ_2 and λ_3) are a complex conjugate pair, we use the identity given in the problem to first write

$$p_{ii}^{(n)} = a_1(1)^n + \left(-\frac{1}{\sqrt{3}}\right)^n \left(a_2 e^{-in\pi/6} + a_3 e^{in\pi/6}\right),$$

and then (as done in **Example 1.1.6**) expand as a linear combination of cosine and sine to obtain

$$p_{ii}^{(n)} = a_1(1)^n + \left(-\frac{1}{\sqrt{3}}\right)^n \left(\tilde{a}_2 \cos\left(\frac{n\pi}{6}\right) + \tilde{a}_3 \sin\left(\frac{n\pi}{6}\right)\right).$$

We determine the constants $a_1, \tilde{a}_2, \tilde{a}_3$ using three values of $p_{ii}^{(n)}$ computable directly from P :

- $n = 0$: trivially $p_{ii}^{(0)} = 1$.
- $n = 1$: directly from the diagonal of P , $p_{ii}^{(1)} = 0$.
- $n = 2$: $p_{ii}^{(2)} = \sum_k p_{ik} p_{ki} = p_{i,i+1} p_{i+1,i} + p_{i,i-1} p_{i-1,i} = \frac{2}{3} \cdot \frac{1}{3} + \frac{1}{3} \cdot \frac{2}{3} = \frac{4}{9}$.

Substituting these into the general form gives a linear system

$$n = 0 : \quad 1 = a_1 + \tilde{a}_2,$$

$$n = 1 : \quad 0 = a_1 - \frac{1}{\sqrt{3}} \left(\frac{\sqrt{3}}{2} \tilde{a}_2 + \frac{1}{2} \tilde{a}_3 \right) = a_1 - \frac{1}{2} \tilde{a}_2 - \frac{1}{2\sqrt{3}} \tilde{a}_3,$$

$$n = 2 : \quad \frac{4}{9} = a_1 + \frac{1}{3} \left(\frac{1}{2} \tilde{a}_2 + \frac{\sqrt{3}}{2} \tilde{a}_3 \right) = a_1 + \frac{1}{6} \tilde{a}_2 + \frac{1}{2\sqrt{3}} \tilde{a}_3.$$

Solving the linear system gives $a_1 = \frac{1}{3}, \tilde{a}_2 = \frac{2}{3}, \tilde{a}_3 = 0$, allowing us to write the final probability as

$$p_{ii}^{(n)} = \frac{1}{3} + \frac{2}{3} \left(-\frac{1}{\sqrt{3}}\right)^n \cos\left(\frac{n\pi}{6}\right).$$